Research Design, Biostatistics, and Literature Evaluation

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Learning Objectives

1. Identify factors influencing the conduct of essential critical care research.
2. Evaluate the appropriateness of various statistical tests for a set of data.
3. Apply concepts of research design and analysis to clinical care.

Abbreviations in This Chapter

4F-PCC	Four-factor prothrombin complex concentrate
ARDS	Acute respiratory distress syndrome
ARR	Absolute risk reduction
CI	Confidence interval
CRP	C-reactive protein
ICU	Intensive care unit
IRB	Institutional review board
NMBA	Neuromuscular blocking agent
NPV	Negative predictive value
OR	Odds ratio
PPV	Positive predictive value
QI	Quality improvement
RR	Relative risk
RRR	Relative risk reduction

Self-Assessment Questions

Answers and explanations to these questions may be found at the end of this chapter.

1. A clinical trial is being planned to determine the optimal resuscitation fluid for trauma patients. This study will seek to determine whether administering crystalloid fluid, such as lactated Ringer solution, or a transfusion strategy involving packed red blood cells in the trauma field improves survival to hospital discharge. The investigator team has consulted with various ethics scholars to identify relevant issues to be addressed in the trial design. Which issue is most relevant to the ethical conduct of this study?
   A. Treatment blinding.
   B. Uninformative study population.
   C. Consent obtained from injured subjects.
   D. Design as a noninferiority trial.

2. The investigators of a new study are seeking to identify the optimal end point or study population to test the effectiveness of a novel drug compound for septic shock. The novel drug compound is a recombinant protein that mediates the inflammatory cascade of sepsis and has had impressive results for all etiologies of septic shock in preclinical animal studies. Historical estimates of septic shock mortality are 35%–40% in previous studies from the past 10 years; this guides the investigators to calculate a required sample size of ~800 patients to achieve 80% power. A more recent study of septic shock reported 28-day mortality to be 20% in the standard treatment arm. Which best describes the rationale for selecting a study population or primary end point?
   A. The study should limit study inclusion to patients without a high level of comorbid conditions at baseline to limit confounders.
   B. The study should expand the target population to include patients with sepsis, not just septic shock, to show a 28-day mortality benefit.
   C. The sample size of the new study will need to be larger than earlier studies to increase likelihood of obtaining power.
   D. The study should limit study inclusion to patients with septic shock caused by pneumonia to test a relevant study population.

3. A quality improvement (QI) initiative is implemented to improve antimicrobial dosing for patients with septic shock presenting to the emergency department. As the pharmacy representative, you have worked with the pharmacy operations team to ensure that an appropriate selection of antibacterial agents and doses is available in the automated drug-dispensing machines. Which factor will be most important in showing the effectiveness of this QI initiative?
   A. Obtaining informed consent for participation in the QI initiative.
   B. Identifying patients with septic shock in triage.
   C. Creating a community advertising campaign to bring awareness to the initiative.
   D. Determining the social value of the QI initiative.

4. An epidemiologic study seeks to determine the impact of adverse drug events in the intensive care unit (ICU) on patient outcomes, which have an estimated incidence of 52%. Which would be the best approach to conducting this study?
   A. Randomized controlled trial with a test for continuous variables to determine the difference in outcomes.
   B. Retrospective case-control study with a test for proportions to determine the difference in outcomes.
   C. Prospective observational study with survival analysis to determine the difference between cohorts.
   D. Retrospective cohort study with a test for proportions to determine the difference in outcomes.

5. A case-control study is conducted to determine whether proton pump inhibitor (PPI) use is associated with an increased risk of developing *Clostridium difficile* infection (CDI). The final analysis shows the odds ratio (OR) for CDI with PPI exposure to be 1.3 (95% confidence interval [CI], 0.8–1.5). Which best describes the results?
   A. PPI exposure increases the risk of CDI by 130%.
   B. PPI exposure reduces the risk of CDI by 20%.
   C. PPI exposure increases the risk of CDI by 30%.
   D. PPI exposure is not associated with an increased risk of CDI.

6. A systematic review evaluated the effect of albumin for fluid resuscitation. A meta-analysis that evaluated the effect of albumin use compared with normal saline on 28-day mortality reported a combined OR of 0.45 (95% CI, 0.3–0.75) for the treatment of hypovolemic shock caused by trauma. For the treatment of septic shock, albumin compared with normal saline resulted in a combined OR of 1.1 (95% CI, 0.98–1.21) when evaluating 28-day mortality. Which best represents the findings of the review?
   A. Albumin increased mortality in trauma but did not affect survival in the treatment of septic shock.
   B. Albumin increased survival in trauma but did not affect survival in the treatment of septic shock.
   C. Albumin did not affect survival in the treatment of hypovolemic shock caused by trauma but improved survival in the treatment of septic shock.
   D. Albumin did not affect survival in the treatment of hypovolemic shock caused by trauma or septic shock.

7. A critical care pharmacist is faced with an acute drug shortage in which no furosemide is available for immediate use in patient care. During patient care rounds in the ICU, the decision is made to implement a fluid-conservative strategy for a patient with acute respiratory distress syndrome (ARDS) (central venous pressure goal less than 4 mm Hg). The critical care pharmacist procures an allotment of bumetanide. Which statement best describes the course of action for this patient?
   A. The pharmacist uses an understanding of the medical literature and experiential knowledge to develop a titration scheme using bumetanide to achieve a central venous pressure of less than 4 mm Hg.
   B. The pharmacist uses an understanding of research trial design and experiential knowledge to develop a titration scheme using bumetanide to achieve a central venous pressure of less than 4 mm Hg.
   C. The pharmacist uses a friendly rapport to convince the nephrologist to treat this patient with hemodialysis to achieve a goal central venous pressure of less than 4 mm Hg.
   D. The pharmacist uses an understanding of research ethics to obtain informed consent from the patient's surrogate for treatment with bumetanide.

8. In a study of ARDS, patients are treated with a neuromuscular blocking agent (NMBA) or placebo to determine whether administering a NMBA within the first 48 hours of presentation improves 28- and 90-day survival. The study enrolled an unequal distribution of patients, with more patients having severe ARDS than moderate and mild ARDS. The post hoc analysis of the results showed a survival benefit with NMBA administration for the patients with severe ARDS. Which rationale best describes why NMBAs should not be administered to patients with mild to moderate ARDS?

   A. Patients with mild ARDS have a lower mortality rate and are therefore less likely to benefit from the test treatment.
   B. Patients with mild and moderate ARDS are inherently different from patients with severe ARDS because of their etiology and presentation.
   C. NMBAs are periodically on shortage from the manufacturer and need to be prioritized for necessary medical indications.
   D. The end points selected do not carry sufficient social value to warrant treatment.

BPS Critical Care Pharmacy Examination Content Outline

This chapter covers the following sections of the Critical Care Pharmacy Examination Content Outline:

1. Domain 3: Professional Practice
    a. Task B: 1–3

I. INTRODUCTION

A. Epidemiology of Critical Care in the United States – Why continued research is essential to improving the delivery of care to patients
   1. 10.1%–28.5% of all hospital inpatients receive care in an intensive care unit (ICU), approximating 5.7 million adults. Estimated mortality for ICU care is 20%–40% for common critical care syndromes (Crit Care Med 2012;40:1072-9).
   2. 16.9%–38.4% of total hospital costs are spent on critical care services, approximating $121–$263 billion.
      a. 5.2%–11.2% of national health care expenditures
      b. 1% of gross domestic product

B. Why Pharmacists Need to Understand the Fundamentals of Research Practice, Trial Design, and Literature Evaluation – High rates of morbidity and mortality necessitate efficient and efficacious decision-making on the part of caregivers.

C. Necessity of Clinical Research in Critical Care – To optimize patient outcomes and minimize potential patient harm while providing efficient stewardship of finite resources

D. Synthesizing Medical Knowledge with Experiential Knowledge and Pathophysiologic Reasoning – Essential to creating patient-specific therapy care plans

II. BIOETHICS

A. The Belmont Report – Originally created as a result of the National Research Act of 1974; outlines the fundamental ethical principles for the conduct of clinical research
   1. Respect for individuals dictates that each research participant be treated with respect for his or her dignity and autonomy. As such, informed consent shall be obtained from research participants or their surrogates.
   2. The principle of justice requires that investigators recruit research subjects in a manner that allows equal access to participation for all populations that may potentially benefit from the research endeavor.
   3. Beneficence requires research investigators to ensure that risks are minimized and benefits maximized for research participants.

B. A framework for the ethical conduct of clinical research includes seven requirements: (1) social value, (2) scientific validity, (3) fair selection of research participants, (4) favorable risk-benefit ratio, (5) independent review, (6) informed consent, and (7) respect for enrolled participants (JAMA 2000;283:2701-11).

C. Equipoise – Must be present for the conduct of a clinical trial. Equipoise is defined as the state of uncertainty between treatments A and B for a given population of subjects with a predefined disease and/or syndrome. Once the balance of uncertainty between treatments A and B is disturbed such that one treatment is believed to be superior, the risk-benefit ratio is altered such that treatment may not be beneficial to the individual research subject. Example: a proposal for clinical research ethics in critically ill patients during pandemics (Crit Care Med 2010;38:e138-142)

D. The Common Rule – Created in 1991 as uniform ethical standards for research after a series of exposures of unethical research misconduct, such as the Tuskegee syphilis study. This rule defined human participant research, developed the process of informed consent, and established the requirement of institutional review boards (IRBs) (JAMA 2017;317:1521-2).
   1. The Common Rule has undergone major changes, with most implementation occurring in 2019 (Chest 2019;155:272-8). These included the following:
      a. New categories and definitions of exempt or excluded research studies
      b. Allowance of a single IRB for multi-institutional studies
      c. Exclusion of public health surveillance as research
      d. Requirement of concise, focused consent forms and online posting of consent as applicable
      e. Allowance of broad consent for identifiable biospecimens in future unspecified research studies

III. PRACTICAL CHALLENGES TO CRITICAL CARE RESEARCH

A. Institutional Review Board (IRB)
   1. IRB of record versus institutional versus independent versus relying IRB
   2. Important considerations for a priori planning
      a. Differences: Processes, time, cost, forms, etc.
      b. Central IRB

B. Research Subject Recruitment – To maximize external validity, research subjects recruited for participation in a clinical trial should be representative of the general population of patients with the disease or syndrome.
   1. Critical care is exemplified by the provision of supportive care for the treatment of diseases and syndromes.
      a. Diseases are characterized by the specific test used to identify the pathophysiologic process.
      b. Syndromes are often identified by the presence of a constellation of signs and symptoms that suggest the presence of a disease that exists across a spectrum of severity.
   2. Recognizing the attendant syndrome is critical for the timely provision of therapy and, in research, subject recruitment.
      a. Heterogeneity in syndromes challenges the ability to identify subjects with similar severity or probability for similar clinical outcomes for participation in clinical research.
      b. In addition, heterogeneity challenges the ability to interpret the results from dissimilar populations, even though they may have a single syndrome in common.
      c. Discrepancies between results from basic and early clinical studies versus adequately powered controlled trials show that surrogate outcome measures (e.g., organ function) or biomarkers may not reliably predict clinical benefit (Am J Respir Crit Care Med 2015;191:1367-73).
   3. Unique to critical care clinical research is the need to enroll subjects similar in acuity or at a similar stage in the process of their syndrome to allow meaningful comparisons (e.g., N Engl J Med 2010;363:1107-16).
   4. Use of non-standardized definitions and lack of power complicate data synthesis and meta-analysis (Am J Respir Crit Care Med 2014;189:1469-78).
   5. Selection of end points should be based, in part, on the ethical values outlined previously. End points should provide social value and have scientific validity. A growing area of emphasis in research is on the need to design patient-centered studies (JAMA 2012;307:1583-4). (Example: Positive vs. negative neurological outcome, rather than mean change in modified Rankin Score.)

C. Informed Consent
   1. Necessity depends on many factors (e.g., IRB or trial design)
   2. 6th–8th grade literacy level
   3. Because of the acute nature of critical illness, it may be singularly difficult to obtain informed consent within a short time.
   4. Obtaining informed consent is particularly challenging for critically ill patients, primarily because many critically ill patients lack decisional capacity because of the acute nature of their illness and/or the effect of medications.
   5. Research that involves subjects with impaired consent within the ICU should include plans for assessment of capacity. The University of California, San Diego Brief Assessment of Capacity to Consent (UBACC) is a 10-item scale that assesses capacity for enrollment into clinical research (Arch Gen Psychiatry 2007;64:966-74).
   6. Surrogates and family members are recognized as having authority to provide consent on the behalf of patients, often termed "legally authorized representative," despite the ambiguous legal standing on this issue in several states.
      a. It is important to realize that underlying motives for proxy consent may include the belief that participating in the research protocol will lead to improved care.
      b. Important to acknowledge that patients and surrogates entrust clinical researchers to act in their best interests
      c. Research supports that surrogate consent and patient preferences agree most of the time, but discrepancies may occur in studies considered higher risk (Chest 2001;119:603-12; Crit Care Med 2012;40:2590-4).
   7. One area of growing utilization is digital consent (even video consent). Digitalized consent has the ability to increase comprehension, particularly in those with poor literacy or a lower socioeconomic status (JAMA Cardiol 2020;5:845-7; JAMA 2015;313:463-4; EPMA J 2018;9:225-34).

D. Waiver of Informed Consent
   1. In select circumstances, informed consent may be waived by the IRB.
   2. Of note, differences exist in the guidance between the Department of Health and Human Services and the U.S. Food and Drug Administration (FDA) for waiver of informed consent. In many circumstances, local IRBs will follow the Common Rule as the minimum standard. Further, restrictions regarding the use of a waiver of informed consent vary by country and culture (Theor Med Bioeth 2016;37:441-6; Bioethics 1999;13:249-55).
   3. Waiver of informed consent is typically granted for any of the following circumstances:
      a. Research that is deemed of minimal risk to the participant, does not adversely affect the welfare of the subject, and could not otherwise be practicably carried out
      b. Research that is carried out to evaluate public benefits or service programs
      c. Research in emergencies when consent is not possible, delays lifesaving intervention, or involves standard-of-care therapy.
         i. An example is a study testing the hypothesis of whether drug A is noninferior to drug B for status epilepticus (N Engl J Med 2012;366:591-600).
         ii. A trial conducted in France that investigated the timing of initiation of renal replacement therapy did not require written informed consent because the standard of care was included in both treatment arms (N Engl J Med 2016;375:122-33).
   4. Controversy exists regarding waiver of consent for quality improvement (QI) research. Currently, QI research is subject to the interpretation of local IRBs, as mentioned previously (N Engl J Med 2015;372:855-62).
      a. Keystone ICU project was a joint collaboration between investigators at Johns Hopkins University and the state of Michigan to reduce central line -associated bloodstream infections in ICU patients using a checklist (N Engl J Med 2006;355:2725-32).

i. Study was classified as exempt from IRB review and informed consent because of the study's QI nature.
ii. After a complaint, the Office for Human Research Protections of the Department of Health and Human Services originally faulted the investigators for not obtaining informed consent. Subsequently, the Office for Human Research Protections reversed course and classified the project as a "non-research" activity.

b. Inconsistent interpretation by local IRBs, in the absence of clear guidance from the Office for Human Research Protections, contributes to confusion regarding the necessity of informed consent in QI initiatives (Jt Comm J Qual Patient Saf 2008;34:349-53).

E. Community Consent
1. Occasionally, critical care research will need to be conducted in the general community. Obtaining consent in this scenario would be impossible, given the medical condition of the research subject. Example: intramuscular therapy compared with intravenous therapy for prehospital status epilepticus (N Engl J Med 2012;366:591-600). In this study of prehospital emergency care, informed consent would be impossible to obtain in the field. Therefore, IRBs engaged local community consultation, according to regulations, and provided approval.
2. Guidance exists for investigators to inform the community and community leaders before undertaking the research endeavor.
3. Approval for this type of research is required from local IRBs.
4. This type of approval falls under Exception from Informed Consent
   a. Not to be confused with Waiver of Informed Consent (Prehosp Disaster Med 2019;34:111-3)
   b. Consider the affected community (e.g., culture, vulnerabilities, etc.) (Prehosp Disaster Med 2018;33:457-8)

F. Social Determinants of Health
1. The conditions in which people are born, grow, live, work, and age
   a. Education level/income/occupation/recreational opportunities
   b. Physical environment (e.g., housing, utilities availability, neighborhood, safe drinking water, clean air, transportation, etc.)
   c. Gender inequity
   d. Racial segregation
   e. Food insecurity/nutritious choices
   f. Early childhood experiences/development
   g. Crime or violent behavior exposure
2. Money, power, resources
   a. Global, national, local stratifications and inequities
   b. Complex interplay (poor health/education↔employment↔income↔healthcare/nutrition↔stress↔substance abuse/unhealthy habits)
3. May have a higher impact on health than healthcare

Practice Case

1. You are working with a multicenter group to develop a cluster-crossover randomized trial. Patients will be recruited at 12 large academic medical centers in eight metropolitan cities. The study will involve a comparison of intravenous haloperidol with intravenous droperidol for acute agitation in the emergency department. Which of the following would NOT qualify as an essential practical consideration for this proposed study design?
   A. Informed consent - unclear whether waiver of informed consent would be appropriate because both agents may be considered current standard of care or whether community consent might be more appropriate.
   B. Social determinants of health - are some patients more likely to be enrolled in study depending on local perceptions of etiologies of agitation (e.g., drug use, psychological disorders)
   C. Subject recruitment - important to ensure that homogeneous populations are enrolled despite institutional and geographic differences that could significantly affect results.
   D. Blinding - unnecessary bias is introduced into the study due to the inability to blind patients and providers to drug selection.

IV. STUDY DESIGN (FIGURES 1 AND 2)

A. Randomized Controlled Trial
   1. Hallmark of clinical research. Study design seeks to minimize bias through randomization of subjects, blinding of investigators and all participants, and analytic approach. This should leave at least two groups that differ only in study treatments.
   2. Experimental design to test the effects of an intervention compared with either placebo or the established standard of care (treatment or process of care); allows for description and causality

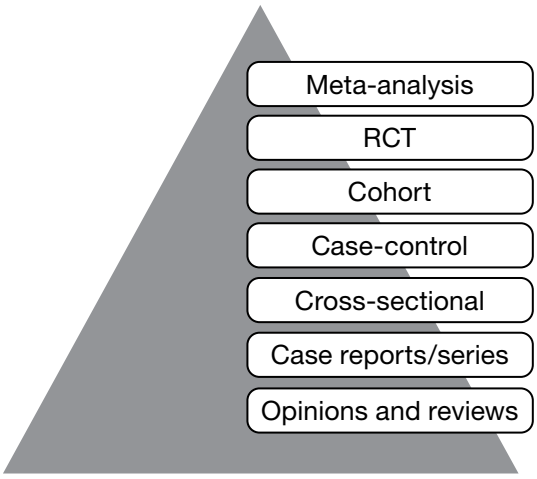


Figure 1. Hierarchy of clinical evidence.

RCT = randomized controlled trial.

Sevransky JE, Checkley W, Martin GS. Critical care trial design and interpretation: a primer. Crit Care Med 2010;38:1882-9.

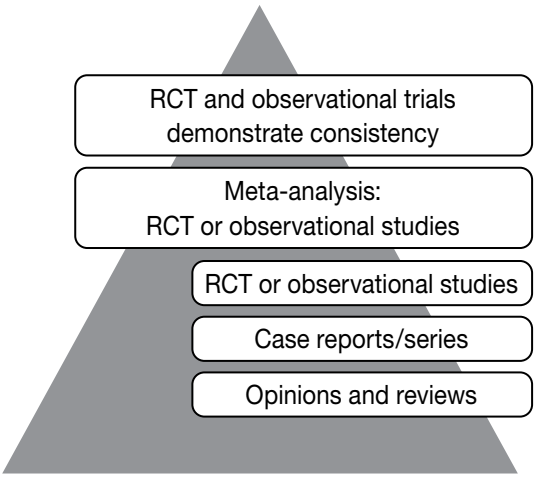


Figure 2. Proposed revision to hierarchy of clinical evidence.

Gershon AS, Jafarzadeh S, Wilson KC, et al. Clinical knowledge from observational studies. Am J Respir Crit Care Med 2018;198:859-67.

3. Preliminary research should exist to suggest that the intervention is based on pathophysiology and repeated outcomes consistent with the existing scientific understanding sufficient to warrant the proposed testing on patients.
4. While historically considered the gold standard for research, many limitations exist to the application and interpretation of randomized controlled trials in the ICU (Front Immunol. 2018;9:1502).
    a. Heterogeneous population: The assumption that all patients would benefit from a treatment is unlikely.
    b. Gaps in understanding: Prognostic markers and definitions are often limited, raising concern for inappropriate inclusion and/or exclusion.
    c. Therapeutic optimization often unknown: Preclinical data is often highly different from the population studied (healthy humans or rodent models) limiting confidence in extrapolation.
    d. Clinical equipoise: Uncertainty surrounding the intervention in question may be subjective, and opinions on what would be considered the standard of care may differ at local, national, and international levels in the critically ill. An ethical approach to the comparator arm must therefore be carefully considered (Circulation 2007;115:1164-9).
5. Trial enrichment is a mechanism in which a specific subgroup of a population who have higher likelihood of a response to an intervention are selected over those who may not (Nat Rev Nephrol 2020;16:20-31).
    a. Prognostic enrichment selects for patients with a higher likelihood of disease-related interest (i.e., mortality)
    b. Predictive enrichment selects for those patients who are more likely to respond to a treatment based on underlying biologic mechanisms or physiology (i.e., inflammatory markers in ARDS)

B. Observational
1. Observation of clinical practice; no intervention is tested
2. Describes associations between phenomena
3. Case-control study
    a. Includes patients who did and did not experience an outcome of interest
    b. Hypothesis generating and retrospective design
    c. Provides cost-effective means to determine the association between the risk factor and the outcome of interest
    d. The case group that experienced the outcome are compared with the control group that did not experience the outcome to identify the differences and risk factors (i.e., exposures) for developing the outcome of interest.
    e. Potential for selection bias and confounding because study proceeds from effect (subjects selected based on outcome of interest) to cause of effect
    f. Cases and controls are representative of the population with the disease and should be chosen in a way to minimize selection bias. Controls are randomly selected out of the same population from which the cases arose.
4. Cohort study
    a. Includes patients who did and did not experience an exposure of interest. May be prospective or retrospective in design
    b. Observational study of a given population over a given time to determine the relationship between exposure and outcome of interest
    c. Describes the natural progression of a disease or syndrome
    d. Example: Delirium, defined as a positive Confusion Assessment Method for the ICU (CAM-ICU) examination, is an independent predictor of mortality in mechanically ventilated patients (JAMA 2004;291:1753-62).

i. Cohort of mechanically ventilated patients observed during ICU stay for presence of delirium
ii. Determination of outcome: ICU mortality, 6-month mortality
iii. Regression models to determine whether delirium is predictive of 6-month mortality, controlled for covariates

5. Case reports/case series
   a. Describes the experience in treating a single patient or the cumulative experience in treating a series of patients
   b. Typically novel or rare patient population or intervention
6. Propensity score matching
   a. This method in theory allows for design and analysis of observational studies to mimic aspects of a randomized controlled trial.
   b. The propensity score itself distributes observed baseline covariates similarly between treated and untreated subjects to create a certain number of matched sets.
   c. Caution should be exercised as propensity scores ensure balance only in observed covariates, where randomization ensures balance between both observed and unobserved covariates.
7. Challenges with observational studies
   a. Missing data are frequently a problem within observational research and may be classified as missing completely at random, missing at random, or missing not at random (Am J Epidemiol 2012;175:210-7).
      i. Missing completely at random data are truly random and case-dependent and are at decreased risk for introduction of bias
      ii. Missing at random indicates data that are absent and the absence is related to other patient data (e.g., correlation with age), and therefore may increase risk of bias.
      iii. Missing not at random indicates data that are absent, but the absence is not related to one of the above.
   b. There are multiple ways to handle missing data (e.g., imputation), but the method should be defined a priori given that significant amounts of missing data may introduce bias.
   c. Confounding variables must be handled in a manner that can be controlled during analysis (i.e., regression models).

C. Meta-analysis: A systematic approach to the identification and abstracting of critical information from research reports. Outcomes from a meta-analysis may include a more precise estimate of the treatment effect or risk factor for disease than the individual studies that contributed to the pooled analysis.
   1. Provides examination of heterogeneity or variability in responses
   2. May be applied even when the included studies are small and substantial variation exists in the issues studied, research methods, study subjects, and other factors that may affect the overall findings
   3. Many meta-analyses combine results into a best estimate with statistical confidence bounds meant to summarize what is known about the clinical problem in question.
   4. Pooled results may incorporate the biases of individual studies and embody new sources of bias (e.g., publication bias).

D. Adaptive trials: Increasing attention has been given to the use of adaptive trial designs. Such trials seek to increase flexibility in clinical trials by allowing for trial modification. Prespecified rules dictate modifications upon scheduled interim evaluations of the data as the trial is ongoing (BMC Med 2018;16:29).
   1. Examples of adaptive designs include: Continual reassessment method; group-sequential method; sample size re-estimation; multi-arm, multi-stage population enrichment; biomarker adaptive; adaptive randomization; adaptive dose-ranging; and seamless phase I/II or II/II trials.

2. Adaptive trials seek to increase efficiency of any type of prospective trial and to minimize potential harm to patients.
3. Challenges to adaptive trials include the complexity of statistical interpretation, lack of knowledge of the scientific community about these designs, and difficulty in communication of the results.

E. Cluster randomized trials
1. Randomize groups (clusters) of subjects to either control or intervention groups
2. Require a large sample size given the intracluster correlation coefficient and resulting design effect
3. Useful in system or group-level intervention (i.e., medication diluent change for entire ICU), when individual randomization is not possible, and to prevent contamination, or the phenomenon which occurs when providers or subjects learn about the intervention during the study period and start adopting it as standard of care, whether within the treatment group or not
4. May result in recruitment bias, more baseline imbalance between groups, loss of entire clusters, or inappropriate analyses (Arztebl Int. 2018;115:163-8).

F. Pharmacokinetic and Pharmacodynamic trials
1. Studies designed to evaluate effects of unique disease states on clinically relevant pharmacokinetic and pharmacodynamic parameters
2. Trial designs can be prospective based on predefined sampling strategies or retrospective based in routine clinical care laboratory samples (Clin Pharmacokinet 2015;7:783-95)
3. Large study size not needed to define properties in unique patient populations
4. Population based pharmacokinetic and pharamacodynamic studies can help define novel clinical dosing strategies in unique patient populations

G. Noninferiority trial
1. Noninferiority trials seek to answer whether a competing treatment is no worse than the established standard therapy (JAMA 2015;313:2371-2).
   a. H_0 = treatment B is inferior (worse) than treatment A.
   b. H_1 = treatment A is no worse than treatment B by X margin.
2. Seeks to establish utility by showing similar effectiveness while improving safety or reducing treatment burden (e.g., costs, inconvenience, labor, etc.) (JAMA 2012;308:2605-11).
3. Analytic approach to a noninferiority trial
   a. What is an acceptable threshold for "noninferiority"?
      i. The FDA provides guidance for non-inferiority thresholds:
         (a) Establish the smallest plausible benefit of the existing standard therapy.
         (b) Insist on some preservation of the treatment effect of the standard therapy.
         (c) The FDA recommends that 50% of the standard treatment effect be preserved when evaluating mortality for thrombolytic trials.
      ii. Maximum allowable negative outcome
   b. Threats to meaningful comparisons
      i. Effect of standard treatment was not preserved. (e.g., heparin infusion does not achieve therapeutic anticoagulation)
      ii. Intention-to-treat analysis: Suboptimal administration of the standard treatment

H. Study Design Considerations
   1. Incidence versus prevalence
      a. Incidence: Measures the occurrence of a disease (or event) over a time interval
      b. Prevalence: Measures the occurrence of a disease (or event) at a fixed point in time
   2. Validity
      a. Internal
         i. Does the study design adequately test the hypothesis?
         ii. Are the study methods sound?
      b. External
         i. Is the study population representative of the clinical setting?
         ii. Are the study findings generalizable outside the study setting?
         iii. Can the study be replicated in clinical practice?
   3. Bias: Systematic error leading to an estimate of association in the study population that varies from the source population (Am J Health Syst Pharm 2008;65:2159-68)
      a. Selection bias: Systematic selection of subjects that leads to an imbalance, or an advantage, in favor of one cohort over the other
      b. Observation bias: Observers (research team) are aware of the research purpose and allow this knowledge to influence interpretation of results.
      c. Recall bias: Methodological error that is introduced in survey research when the participant is asked to provide recall of a past event
      d. Misclassification bias: Inappropriately categorizing a group of patients with, or without, the disease/syndrome
      e. Immortal time bias: Occurs when populations studied include an exposed and unexposed group without a predefined time-zero. This results in a delay before a subject is considered treated. Given that the subjects must survive up until the time of first exposure, the time before exposure is considered immortal time (Am J Respir Crit Care Med 2003;168:49-53).
      f. Measurement bias: When information collected for use as a study variable is inaccurate
      g. Confounding variables: Extraneous variables that influence both the dependent and the independent variable, affecting how the overall result can be interpreted
         i. Residual confounding occurs when unmeasured variables may affect overall outcome results. This can only be avoided with randomization (Int J Public Health 2010;55:701-3).
         ii. Confounding by indication occurs when a contributor to the outcome is present in those at high risk and is an indication for intervention. This results in care differences between the exposed and nonexposed groups that may be based on differences in indication for exposure (Pharmacoepidemiol Drug Saf 2003;12:551-8).
      h. Bias can be accounted for with sufficient planning for design, data collection, and analysis and can potentially be minimized (i.e., "effect modification").
   4. Sensitivity analysis: These may be used to determine robustness of study findings in different analysis methods or in the presence of unmeasured confounding. They may also be used to detect misclassification (Am J Respir Crit Care Med 2018;198:859-67).
   5. Approaches to control for bias:
      a. Study design: Randomization, matching
      b. Analysis: Multivariable analysis, matching potential confounding variables with a controlled cohort

Practice Case

Questions 2 and 3 pertain to the following case.
You are a critical care pharmacist in the medical ICU and wish to retrospectively study the safety and efficacy of four-factor prothrombin complex concentrate (4F-PCC) for coagulopathy correction in patients with end-stage liver disease. You decide on a primary outcome of in-hospital mortality and secondary outcomes of INR correction, viscoelastic test correction, and thrombotic events.

2. Which is likely the most appropriate trial design for this study?
   A. Case-control.
   B. Cohort.
   C. Case series.
   D. Pharmacokinetic trial.

3. Which would create the highest risk of bias in this study?
   A. Selection bias – typically only patients who are bleeding receive 4F-PCC at your institution.
   B. Measurement bias – multiple students and a pharmacy resident collected data for this study.
   C. Misclassification bias – classification of patients as having end-stage liver disease requires manual chart review.
   D. Observation bias – there was no blinding of those who did or did not receive 4F-PCC.

V. STATISTICAL ANALYSIS

A. Statistical analyses can be categorized into one of two main classifications: Bayesian or central tendency (frequentist)
   1. Bayesian analysis allows for incorporation of prior knowledge into current study and calculates the probability of benefit.
   2. Frequentist statistics data are presented using descriptive statistics such as the mean, variance, standard deviation, median, and interquartile range.
   3. Interpretation of studies may be different based on whether frequentist or Bayesian analyses are selected (JAMA 2019;321:654-64; Am J Respir Crit Care Med 2020;201:423-9).

B. Hypothesis Testing – Determining whether an observation or outcome was caused by chance results in either rejecting or accepting (failing to reject) a null hypothesis (H_0)
   1. Rejecting the null hypothesis results in accepting the alternative hypothesis (H_1)
   2. Null and alternative hypotheses are built according to study aim and design
      a. Superiority trial example (N Engl J Med 2023; 388:499-510).
         i. H_0 = no difference in mortality exists between a restrictive versus liberal fluid strategy for sepsis-induced hypotension
         ii. H_1 = a restrictive fluid strategy would lead to lower mortality than liberal fluid strategy for sepsis-induced hypotension
         iii. The study found no significant difference in mortality between groups, thereby resulting in a failure to reject the null hypothesis.

b. Type I error (alpha [α] error): To reject the H_0 when, in fact, it is true. Decisional threshold to reject/not reject the H_0 is conventionally set at $\alpha = 0.05$. The α value represents the likelihood that a type I error will be made. An α set at 0.05 means that the H_1 will erroneously be accepted 1 in 20 times.
   i. Two-tailed tests distribute half of the α value at the far left hand and far right hand ends of the distribution
   ii. One-tailed tests distribute all of the α value at the far left hand or far right hand ends of the distribution
c. Type II error (beta [β] error): Fail to reject the H_0 when, in fact, there is a difference between groups (i.e., the H_1 is true). Decisional threshold to set β at 0.20.
d. Power (1-β): The ability to detect a difference if one truly exists. Contingent on sample size. However, this is an estimate and may be inaccurate (usually based on previous literature).
   i. Sample size is the number of subjects or samples collected or assessed within a study and is influenced by the following:
      (a) Treatment effect, or difference within the predicted outcomes that is considered to be clinically significant (i.e., difference in control vs. treatment group mortality rate)
      (b) The variability in the predicted outcomes (i.e., the variability in baseline mortality)
      (c) What is deemed to be an acceptable rate of type I error (α)
      (d) Type II error rate (β)
   ii. Typically the greater the effect and the lower the variability, the smaller the sample size required
   iii. Decreasing the rate of type I error (α) and/or increasing the power (decreasing β) increases the necessary sample size.

Table 1. Simple Statistical Test Selection

Variable	Two Samples		>Two Samples	
	Independent	**Related**	**Independent**	**Related**
Nominal	Chi-square or Fisher exact	McNemar test	Chi-square	Cochran Q
Ordinal	Wilcoxon rank sum Mann-Whitney *U*	Wilcoxon signed rank	Kruskal-Wallis (MCP)	Friedman ANOVA
Continuous with no covariates	Student t-test Unequal variance t-test	Paired t-test	1-way ANOVA (MCP)	Repeated measures ANOVA
Continuous with one covariate	ANCOVA	2-way repeated measures ANOVA	2-way ANOVA (MCP)	2-way repeated measures ANOVA

ANCOVA = analysis of covariance; ANOVA = analysis of variance; MCP = multiple comparison procedures.

C. Types of Data and Corresponding Tests (Table 1)
   1. Continuous
      a. Counting, chronological order (e.g., 7.0, 7.1, 7.2, 7.3)
      b. Sample tests: t-test, Wilcoxon rank sum/Mann-Whitney *U*
   2. Categorical
      a. Nominal
         i. Categories or groups to which data are designated (e.g., race, sex, hypertension, heart failure)
         ii. Sample tests: Chi-square, Fisher exact
      b. Ordinal
         i. Categorical data that include an order (e.g., Likert scale)
         ii. Sample tests: Wilcoxon signed rank test, Wilcoxon rank sum test, Kruskal-Wallis 1-way test

3. Descriptive statistics
    a. Measures of central tendency
    b. Mean: Used for continuous and normally distributed data (parametric), sensitive to outliers. Typically presented in conjunction with standard deviation.
    c. Median: 50th percentile, used for ordinal data or continuous and non-normally distributed data (nonparametric). Typically presented in conjunction with interquartile range.
    d. Mode: Value occurring most frequently in a distribution
    e. Standard deviation
        i. Used for continuous, parametric data
        ii. Describes variability around the mean
        iii. Approximately 95% of the data fall within two standard deviations of the mean
    f. Range: Describes spread of data, minimum to maximum values
    g. Percentiles: The value where a percentage of values falls below. Example: The 75th percentile is where 75% of values are smaller.
    h. Interquartile range: Describes the values between the 25th and 75th percentiles
        i. Used for continuous, nonparametric data
4. Paired data
    a. There are circumstances in which matching will occur naturally between data points
        i. Duplicate measurements on one sample
        ii. Sequential measurements, such as a pre-post test
        iii. Cross-over trials
        iv. Matched samples

D. Parametric Continuous Data
    1. Two independent groups: Student t-test
    2. Three or more groups: Analysis of variance (ANOVA)

E. Nonparametric Continuous Data
    1. Two independent groups: Wilcoxon rank sum or Mann-Whitney *U*
    2. Three or more groups: Kruskal-Wallis

F. Nominal Data
    1. Two or more categories (e.g., sex, race, treatment groups, CAM-ICU)
    2. Chi-square test
    3. Fisher exact test: More accurate for smaller data sets (fewer than five observations per group)

G. Ordinal Data
    1. Groups represented by scale (e.g., Richmond Agitation-Sedation Scale score, Sequential Organ Failure Assessment score)
    2. Central tendency expressed as median and quartiles
    3. Statistical tests: Wilcoxon rank sum, Mann-Whitney *U*, Kruskal-Wallis

H. Confidence Interval (CI)
    1. A method to describe variability around a point estimate within the study population
    2. CIs offer a more descriptive interpretation of the data.
        a. Magnitude of difference between groups
        b. Range of values (possible spread of point estimates)
        c. A 95% CI describes the interval that would contain the true population point estimate in 95 of 100 random samples.

I. Correlation: Describes the association between two variables
   1. Correlation value (r) contains a range of values from -1 to +1. An r value of -1 or +1 indicates a perfect negative or positive relationship, respectively. Relative to 0, the closer the values are to 1, the stronger the relationship between the two variables.
   2. Pearson: Parametric continuous data
   3. Spearman rank: Nonparametric continuous data or ordinal data

J. Regression Analysis: Describes whether an independent variable can predict an outcome or the causal effect of an exposure on an outcome in the presence of possible confounders (Table 2). Can be used to describe the strength of the association between a predictor variable and a dependent variable
   1. Linear regression
      a. Used when the dependent variable is continuous (e.g., length of stay)
      b. A single predictor variable (continuous or categorical) can be tested in a simple linear regression model.
      c. More than one predictor variable (continuous or categorical) can be tested at a time in a multivariable linear regression model.
   2. Logistic regression
      a. Used when the dependent variable is a categorical variable (e.g., mortality)
      b. A single predictor variable (continuous or categorical) can be tested in a simple logistic regression model.
      c. More than one predictor variable (continuous or categorical) can be tested at a time in a multivariable logistic regression model.
   3. Considerations
      a. Selection of variables may depend on the primary goal of the regression model (predictive vs. explanatory).
         i. Variables for models that examine causal associations should be identified as possible confounders for exposure of interest that may affect the outcome.
         ii. Selecting variables for inclusion solely on the basis of correlation between each independent variable and outcome of interest, p values on univariate analysis, or automatic variable selection (e.g., forward or backward stepwise selection) may result in overfitting or misconceptions about true confounders.
         iii. Avoid colinear variables (i.e., septic shock and vasopressor requirement may be too similar [e.g., multicollinearity])
      b. Goodness of fit may be assessed in multiple ways (R^2, Hosmer-Lemeshow tests, etc.)
         i. Must be critically assessed as goodness of fit may not completely reflect important outliers or may demonstrate overfitting
      c. Regression analyses make several key assumptions, some of which are as follows:
         i. Linear relationships between independent variables and outcome
         ii. No multicollinearity
         iii. Random component of the model is normally distributed
         iv. Appropriate number of variables for study sample size (e.g., at least around 10–20 cases per independent variable in the model)

Table 2. Statistical Regression Model According to Type of Data

Dependent Variables	Independent Variables	Test(s)
Continuous	Continuous	Linear regression
Continuous	Categorical	Linear regression
Categorical, ordinal	Continuous	Logistic regression
Categorical, ordinal	Categorical	Logistic regression

K. Measures of Effect for Categorical Data (Figure 3)
   1. Ratios with 95% CI that are used to compare the incidence of categorical outcomes are interpreted in relation to a reference point (1.0). If the 95% CI surrounding the ratio includes 1.0, the risk, odds, or hazard of the outcome is equally likely in both groups.
   2. Effect estimates are preferred to p values to assess statistical significance. P values are often misused, and effect estimates generally better describe outcomes.
   3. Relative risk (RR)
      a. Incidence of outcome in exposed group/incidence of outcome in control group
   4. Relative risk reduction (RRR)
      a. Relative difference in outcome in exposed group/relative difference in outcome in control group
   5. Absolute risk reduction (ARR)
      a. Absolute difference between outcome in exposed and control groups
   6. Odds ratio (OR)
      a. Odds of outcome occurring in group exposed to risk factor or intervention/odds of outcome occurring in control group
      b. Compares number of events or outcomes between two groups assuming that the incidence remains constant over a fixed time interval
   7. Hazard ratio (HR)
      a. Hazard rate in exposed group/hazard rate in control group
      b. Compares number of events or outcomes between two groups where the incidence varies over a time interval

$$RRR = \frac{ARR}{\left(\frac{C}{C+D}\right)} \qquad ARR = \left(\frac{C}{C+D}\right) - \left(\frac{A}{A+B}\right) \qquad RR = \frac{\left(\frac{A}{A+B}\right)}{\left(\frac{C}{C+D}\right)} \qquad OR = \frac{A+D}{C+D}$$

	Outcome	No Outcome
Exposed	A	B
Unexposed	C	D

Figure 3. Equations for measures of effect.

ARR = absolute risk reduction; OR = odds ratio; RR = risk ratio; RRR = relative risk reduction.

L. Survival Analysis (time-to-event analysis)
   1. Censoring: Adjusts data so that patients are not included (i.e., "censored") in the analysis if they did not experience, or were not observed for, the event
   2. Kaplan-Meier method
      a. Describes the impact of a single predictor variable on the time-to-event between cohorts
      b. Compares the survival times between two cohorts while controlling for a singular predictor variable
      c. Survival curves are typically analyzed using the log-rank test.
      d. Results typically reported as hazard ratio with 95% CI
   3. Cox proportional hazards model
      a. Describes the impact of several predictor variables on the time-to-event
      b. Compares the survival times between two cohorts while controlling for several predictor variables
      c. Results typically reported as hazard ratio with 95% CI

M. Determining Predictive Ability (Figure 4)

$$\text{Sensitivity} = \frac{A}{A + C} \times 100 \qquad \text{PPV} = \frac{A}{A + B} \times 100$$

$$\text{Specificity} = \frac{D}{B + D} \times 100 \qquad \text{NPV} = \frac{D}{B + D} \times 100$$

	Has Disease or Outcome	Does Not Have Disease or Outcome
Positive	A	B
Negative	C	D

Figure 4. Equations for determining predictive ability.
NPV = negative predictive value; PPV = positive predictive value.

1. Sensitivity: Probability of identifying a true positive from a group that is known to be positive
   a. Avoiding false negatives
   b. Example: Proportion of people with disease correctly identified as having the disease through laboratory assay
2. Specificity: Probability of identifying a true negative from a group that is known to be negative
   a. Avoiding false positives
3. Positive predictive value (PPV): Probability of identifying a true positive from a group that may or may not be positive
   a. Avoiding false positives
   b. Example: Proportion of people with positive laboratory assay result who actually have the disease
4. Negative predictive value (NPV): Probability of identifying a true negative from a group that may or may not be negative
   a. Avoiding false negatives

Practice Case

Questions 4–6 pertain to the following case.
The neurosurgery physician group at your institution would like your input in evaluating the usefulness of procalcitonin for distinguishing neurogenic fevers from true infectious etiologies of fever. An additional goal of the study would be to evaluate risk factors and possible confounders for positive and negative procalcitonin values. You decide on a predictive cutoff of 0.5 ng/mL (positive 0.5 mg/mL or greater; negative less than 0.5 ng/mL).

4. Which statistical test would be most appropriate to evaluate any difference in creatinine clearance (milliliters per minute) between patients with positive and negative procalcitonin?
   A. Chi-square.
   B. Wilcoxon signed rank.
   C. Student t-test for parametric data.
   D. Mann Whitney *U* for parametric data.

Practice Case (*continued*)

5. Which statistical test would be most appropriate to determine an independent association between positive procalcitonin and true infection in the presence of possible confounders?
   A. Linear regression.
   B. Logistic regression.
   C. Cox proportional hazards model.
   D. Correlation analysis.

6. You enroll 201 patients who have a true infection and 200 patients who are not infected. Of the 201 with infection, 189 have a positive procalcitonin and 12 have a negative procalcitonin. Of the 200 patients without infection, 100 have a positive procalcitonin and 100 have a negative procalcitonin. Which best describes the sensitivity and specificity of procalcitonin in this population?
   A. 94% sensitivity and 89% specificity.
   B. 65% sensitivity and 89% specificity.
   C. 65% sensitivity and 50% specificity.
   D. 94% sensitivity and 50% specificity.

VI. APPLICATION OF KNOWLEDGE TO PATIENT CARE

A. Integration of Various Types of Knowledge (Table 3)

Table 3. Assessment of Primary Literature for Clinical Application

	Assessment
Study design	• Were the hypotheses and study purpose clearly stated? • Is the study sample representative of the population with the disease/syndrome? • Are the inclusion/exclusion criteria too restrictive? • Did the study meet power? Was a sample size calculation described? • How were blinding and randomization conducted? • Is the study design translatable to clinical practice? • How were the primary and secondary end points defined? Have those definitions been validated in critically ill patients?
Outcomes	• Is the primary outcome scientifically valid and patient-centered? • Are the secondary outcomes clearly described? • How were adverse effects defined and analyzed?
Analysis	• Were the statistical tests appropriate? • How were the data analyzed (intention-to-treat, per-protocol, as-treated)? • How large was the treatment effect? • Will the effect size be duplicated in clinical practice? • Did the author(s) provide an interpretation of the study findings and describe them in the context of the available knowledge?

1. Medical literature
    a. Remaining current with evolving literature is a necessary skill for the critical care clinician.
    b. Knowledge gained from primary literature can be objective, can limit bias, and may be translatable compared with experiential knowledge.
    c. The findings of clinical research can be limited to the conditions of the study and may not easily confer the same benefit in clinical practice.
        i. Number needed to treat (NNT) – Quantifies the anticipated effect of a treatment in a patient population on the basis of study results. A low number signifies an effective treatment; a high number indicates a less effective therapy. NNT = 1/ARR
        ii. Number needed to harm (NNH) – Quantifies the associated harm after exposure to a treatment. A low number signifies a harmful treatment; a high number indicates a safer treatment. NNH = 1/attributable risk
    d. The study results limit the findings attributable to chance but may not fully exclude chance. Remember: Studies are conducted in samples and may or may not be representative of the population.
    e. Fragility index: The number of non-events in the overall sample that need to occur to render a significant result nonsignificant. In the CRICS-TRIGGERSEP trial, an additional three events for the outcome of 90-day mortality would have rendered the trial findings nonsignificant (shifting to a p=0.053), indicating the fragility of the final results (N Engl J Med 2018;378:809-18).
2. Experiential knowledge
    a. Knowledge accumulated through clinical experience and for a sustained period is valuable.
    b. How does an individual patient differ from the patient sample of the study being applied?
    c. When possible, this type of knowledge should be reinforced with scientific evidence. Example: N Engl J Med 2009;361:1925-34
3. Pathophysiologic reasoning
    a. Application of physiologic concepts to drive or reinforce therapeutic choices
    b. Can aid in determining short-term goals. Example: Selecting an initial dose of loop diuretic according to the home medication dose and current therapeutic goal of urinary output
    c. Primary literature on which to base treatment decisions for all clinical scenarios may not exist.

B. Synthesizing the Available Evidence to Develop Treatment Protocols
1. Protocols are intended to synthesize the best available evidence and standardize a series of treatment options to reduce variability and error while maximizing treatment effect. Examples: Sterile technique for central line placement, heparin or insulin infusions, and sepsis bundles.
2. Determining which literature to incorporate and its applicability to the local practice environment is key.
3. Determine the homogeneity of various studies on the same topic (e.g., early goal-directed therapy for septic shock) and confounding variables that influence the interpretation of results. Study design affecting the screening and time to treatment for subjects compared with actual practice, improvements in the processes of usual care over time, etc.

C. Unique Factors That Promote/Impede the Application of Treatment Protocols
1. Resource use
    a. Personnel (e.g., Lancet 2010;375:475-80)
    b. Drug shortages
    c. Fiscal
2. Ease of protocols
    a. Complexity
    b. Familiarity
3. Lack of consensus (e.g., sepsis 1-hour bundle)

Practice Case

Questions 7 and 8 pertain to the following case.
You are reading a study that evaluated the use of hydrocortisone versus placebo for lowering C-reactive protein (CRP) in patients with ARDS at 3 days after enrollment. The authors concluded that they would need to enroll 100 patients to achieve 90% power to find a decrease in CRP of 10 mg/L, assuming a baseline concentration of 35 mg/L and an α level of 0.05. They enroll 50 patients in the hydrocortisone group and 51 patients in the placebo group. The median [interquartile range] CRP concentrations in the intervention and placebo groups are 23 [19–31] and 29 [20–40], respectively ($p<0.05$). The percentages of patients achieving a CRP decrease of at least 10 mg/L in the intervention and placebo groups are 15% and 18%, respectively ($p>0.05$).

7. Which is your biggest concern with interpreting the validity of this study?
   A. The study did not meet statistical power.
   B. The sample size is too small to provide a representative population for the disease state.
   C. The outcome of lowering CRP should not be dichotomized, but only analyzed as a continuous variable.
   D. The outcome of CRP is a surrogate and likely not a good representation of the clinical implications of steroid use in ARDS.

8. Which best describes the NNT for achieving a CRP decrease of at least 10 mg/L in this study?
   A. 34.
   B. 17.
   C. 50.
   D. 6.

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Application of Knowledge to Patient Care

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ANSWERS AND EXPLANATIONS TO PRACTICE CASE QUESTIONS

1. **Answer: D**

This study requires consideration of options A, B, and C. Informed consent may be waived because both drugs may be considered standard of care, but some IRBs at various institutions may require the use of community consent measures; obtaining informed consent directly from the patient or a family member may be impossible due to the acute nature of the intervention. Differences in local perceptions on why patients may be agitated on presentation to the emergency department could lead to heterogeneity in use of the study agents, affecting results downstream. In addition, a multicenter trial creates a risk of differences in patient characteristics during enrollment that may need to be controlled for in the study. Option D is not a valid practical consideration based on the study design. The cluster crossover trial may compare two known standards of care, and the crossover design allows for hospitals/units to serve as their own controls. Risk of bias is never zero, but blinding would not introduce unnecessary bias into this trial's results (Answer D is correct; Answers A, B, and C are incorrect).

2. **Answer: B**

The study goal is to determine whether exposure to 4F-PCC resulted in mortality, or the outcome of interest. This is performed in a cohort of patients who all have end-stage liver disease (Answer B is correct). A case-control study involves structuring study groups by those who did and did not have an outcome of interest in order to determine relevant risk factors leading to that outcome (Answer A is incorrect). A case study would be better suited to a single group without a comparator and is typically used to describe a novel intervention or outcome (Answer C is incorrect). Pharmacokinetic trials typically involve sample collection and evaluation of drug concentrations to describe the kinetics of the drug as opposed to clinical outcomes (Answer D is incorrect).

3. **Answer: A**

If patients who are bleeding more often receive 4F-PCC according to institutional standards or physician preference, there is likely a difference in the severity of illness between the two study groups (those who did and did not receive 4F-PCC), which could lead to selection bias in the result of mortality (Answer A is correct). Although measurement bias is possible because of multiple people collecting data, it is less likely to have as large an impact on the outcomes as long as a data collection plan is implemented and followed (Answer B is incorrect). Misclassification bias is also possible but is less likely to affect outcomes at the same magnitude as selection bias as long as a consistent data collection and chart review plan is followed (Answer C is incorrect). Observation bias would likely be more important in a prospective trial than in a retrospective trial to blind those involved to ensure all other standards of care are followed (Answer D is incorrect).

4. **Answer: C**

Student t-test is appropriate to compare two independent means as a measure of parametric or normally distributed data (Answer C is correct). A chi-square analysis is used for categorical data (Answer A is incorrect). Wilcoxon signed rank is used to compare two related medians as a measure of nonparametric data (Answer B is incorrect). Mann-Whitney *U* is used to compare two independent medians as a measure of nonparametric data (Answer D is incorrect).

5. **Answer: B**

A regression analysis can be used to identify independent associations in the presence of other risk factors for the outcome. Logistic regression is used when the outcome of interest (i.e., patients with positive or negative procalcitonin) is categorical in nature (Answer B is correct). Linear regressions are used similarly to logistic regressions for a continuous dependent variable (Answer A is incorrect). A Cox proportional hazards model is used as a survival analysis or time-to-event analysis, which would not be applicable to the outcome or question in this case (Answer C is incorrect). Although a correlation could determine the strength of association between positive procalcitonin and infection, it would not consider other risk factors that could be confounding the results (Answer D is incorrect).

6. **Answer: D**

The sensitivity is around 94%, and the specificity is 50% (Answer D is correct). The sensitivity is calculated by dividing 189/201, and the specificity is calculated by dividing 100/200. The PPV is around 65%, and the NPV

is around 89% (Answers A, B, and C are incorrect). The PPV is calculated by dividing 189/289, and the NPV is calculated by dividing 100/112.

7. **Answer: D**

When evaluating this study, it is important to consider that the outcome related to a biomarker is simply a surrogate as opposed to a true clinical outcome. This could be problematic or lack validity because other steroids have previously been studied for outcomes such as mechanical ventilator days and mortality (Answer D is correct). The study may not have met the criteria for the original power analysis that was defined by the authors, but it did achieve power to find a statistically significant difference according to the outcome (Answer A is incorrect). Although the sample size is small, it may represent a reasonable population with internal validity to use the information in a limited way, particularly at the institution(s) where it was performed (Answer B is incorrect). The power analysis was created around a continuous outcome, but it is reasonable to present the data in two different ways (in this case, patients who did and did not achieve a CRP decrease of at least 10 mg/L) as long as the reader understands that the study was not powered around this analysis (Answer C is incorrect).

8. **Answer: A**

The NNT is calculated by 1/ARR. The absolute risk reduction is 3%, or 0.03. The NNT is 1/0.03 = 34 (rounding up to the nearest whole number) (Answer A is correct; Answers B, C, and D are incorrect).

ANSWERS AND EXPLANATIONS TO SELF-ASSESSMENT QUESTIONS

1. **Answer: C**
This study seeks to compare two viable treatments of hypovolemic shock in trauma patients. Because subject identification would be trauma patients from the community and treatment would be initiated in the field, potential harm is associated with each treatment, and it is expected that many potential patients would lack decisional capacity at the time of informed consent. Although treatment blinding is an essential component to trial design to limit investigator and clinician bias, it is not an ethical consideration (Answer A is incorrect). Given the incidence of trauma in the general population and its burden on society, treatments to improve the outcomes of trauma patients are necessary and informative (Answer B is incorrect). Acknowledging the limited supply of blood products, a superiority trial is essential to help steward the use of a finite resource (Answer D is incorrect). The issue of informed consent in this study is challenging, but the issue needs to be addressed for the ethical conduct of this study (Answer C is correct). There is precedent for this trial to receive an exception for informed consent requirements under the FDA code of regulations.

2. **Answer: C**
The study seeks to establish the effectiveness of a novel drug compound for septic shock. This is challenging because of the dramatic improvements in 28-day mortality during the past decade. To show the effectiveness of a novel drug compound, the drug should be tested in a representative population of patients with the disease. Excluding patients because of baseline comorbidity would limit external validity (Answer A is incorrect). Including patients with sepsis, which has a lower 28-day mortality rate than septic shock, would not address the issue of an appropriate end point (Answer B is incorrect). Limiting study inclusion to patients with a pneumonia etiology would be inappropriate unless the pharmacology of the novel drug specifically targeted pneumonia pathophysiology (Answer D is incorrect). Answer C is correct as the anticipated incidence of mortality has changed overtime as well as the likely anticipated minimum detectable effect. With all other assumptions remaining equal, the authors will have to increase enrollment to find the effect relative to previous trials.

3. **Answer: B**
A QI study would likely show the effectiveness using a pre/postintervention cohort design. In this type of study, informed consent is generally not required because the treatment is provided to all patients as a standard of care (Answer A is incorrect). A community advertising campaign might improve the delivery of care and improve patient adherence, but it is unnecessary to measure the effectiveness of the QI initiative (Answer C is incorrect). The QI initiative is believed necessary and has therefore been deemed to have intrinsic social value (Answer D is incorrect). To show the effectiveness of the intervention, the syndrome must be recognized before initiating treatment (Answer B is correct).

4. **Answer: C**
To effectively determine the incidence and clinical impact of adverse drug events on clinical outcomes in the ICU, it would be unethical to randomize patients to experience the event (Answer A is incorrect). A retrospective design would not be ideal because of the limitations in data extraction, assignment of events, and interpretation of causality (Answers B and D are incorrect). A prospective observational design would allow the investigator team to identify the incidence of adverse events, sequential outcomes, and determine causality (Answer C is correct).

5. **Answer: D**
A correct interpretation of the results is recognizing that even though the OR suggests an associated increase of 30% in the risk of being exposed to CDI, the 95% CI crosses 1, meaning that the odds of exposure to CDI are as likely to increase the risk as to decrease that risk (Answer D is correct; Answers A–C are incorrect).

6. **Answer: B**
Use of albumin reduced the odds of mortality (and conversely, increased the odds of survival) with an OR of 0.45, and the 95% CIs were all less than 1 in treating hypovolemic shock in trauma. In addition, the OR was 1.1 for septic shock, whereas the 95% CI crossed 1, showing that the odds were as likely that albumin increased mortality as that it reduced it (Answer B is correct). Because the OR and the 95% CI for treating

hypovolemic shock are both less than 1, mortality was decreased with albumin use (Answers A, C, and D are incorrect).

7. **Answer: A**

This patient case provides a practical example of a critical care pharmacist's integration of various types of knowledge to optimize patient care. The FACT trial showed that a fluid-conservative strategy improves ventilator-free days for patients with acute lung injury and ARDS. Furosemide was used in the study to show the outcome benefit. However, the study was testing a treatment strategy, not specifically a drug strategy. Therefore, it can be reasoned that similar treatment outcomes can be shown with similar drugs if the study drug is unavailable – in this case, because of drug shortage. Knowledge of trial design is helpful but not critical to optimizing this patient's therapy with bumetanide (Answer B is incorrect). Hemodialysis is invasive, requires finite resource use, and has an associated morbidity risk (Answer C is incorrect). Because this patient case does not include a broader hypothesis test in a systematic design, this is not a research activity, and informed consent is not required (Answer D is incorrect). A critical care pharmacist, using knowledge of the FACT trial (medical knowledge) together with an understanding and experience with bumetanide therapy (experiential knowledge), can develop a treatment plan (Answer A is correct).

8. **Answer: A**

Acute respiratory distress syndrome is a clinical syndrome with an associated mortality with each progressing phase (mild, moderate, and severe). Although ARDS is also a constellation of findings and pathologic observations, patients with this disease present with a similar finding of severe, refractory hypoxia from a common etiology (Answer B is incorrect). Given the associated mortality of about 45% for severe ARDS, a finite resource should be prioritized for it (in this case, a NMBA) (Answer C is incorrect). Mortality/survival is readily identified as an end point of social value for research design (Answer D is incorrect). Given the relatively low mortality associated with mild ARDS (around 20%) compared with severe ARDS (around 45%), and mortality benefit was seen only in the subgroup of patients with severe ARDS, administering NMBAs would less likely improve survival and more likely increase harm if systematically administered to patients with mild ARDS (Answer A is correct).